PLANNING DOCUMENT

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BTS Cloud Computing

Personas

Persona 1

Name: Jonathan Schmit

Age:38

Occupation: Teacher

Goals & Motivations: Teaching his students well and seeing them pass their years.

Needs / Expectations: Better overview of the grades of his students and their progress.

Frustrations: Other services hadn’t been satisfactory

Persona 2

Name: Jerry Matthias

Age:15

Occupation: Student

Goals & Motivations: Doing well at school so he can graduate and go to University.

Needs / Expectations: Needs a good way to see his grades to assess how he’s doing

Frustrations: Forgets to write his marks down.

Persona 3

Name: Robert Dad

Age:55

Occupation: Father & mechanic

Goals & Motivations: To see his children succeed

Needs / Expectations: Ease of use and good overview

Frustrations: Struggles with the usage of technology

User Stories

Story/Task name	Teacher Grading System	ID	1
		Size	
Description of Customer benefits	As a: **Teacher**		
	I want: To register the grades of my students		
	So that: I have an overview of how my students are doing		
Acceptance Criteria	• **MUST** ○ Grades must be saved in a database ○ Teachers can add tests to their assigned subjects ○ Teachers can grade those tests • **SHOULD** ○ The test can be created without having a grade yet ○ The teacher can correct grades if needed • **COULD** ○ • **WILL NOT** ○ Have a date of when the test will be taken		

- Subtasks:
  - Setup a database to store grades
  - Setup the design for creating tests and grades
  - Program the storage of the test and grades

Story/Task name	Teacher Account	ID	2
		Size	
Description of Customer benefits	As a: **Teacher**		
	I want: my students not to be able to change their grades		
	So that: only I can make changes		

Acceptance Criteria	• **MUST**
	  ○ Have right to modifications of grades
	  ○ Be able to setup classes
	• **SHOULD**
	  ○
	• **COULD**
	  ○ Have their contact details linked to it
	• **WILL NOT**
	  ○		

Subtasks:

- Create the class configuration screen

Story/Task name	Student grades viewing	ID	3
		Size	
Description of Customer benefits	As a: **Student/Parent**		
	I want: my grades to be easily accessible		
	So that: I can focus more on my studies		
Acceptance Criteria	• **MUST** ○ Grades must be viewable within their subjects ○ Show the grades from the • **SHOULD** ○ See the grades from previous tests of the tri-/semester • **COULD** ○ See highest mark and lowest mark of the class ○ View grades from previous years ○ Highlight insufficient grades • **WILL NOT** ○		

Subtasks

- Setup the design of the Grade view
- Show the grades from their respective subject

Story/Task name	Insufficient mark warning	ID	4
		Size	
Description of Customer benefits	As a: **Student**		
	I want: to clearly know if my marks are insufficient		
	So that: I can put more effort into learning for that subject		
Acceptance Criteria	• **MUST** ○ Highlight insufficient marks ○ Have a warning showing the subject and grade • **SHOULD** ○ Tell the student how much they need in their next text • **COULD** ○ Send a notification on an insufficient grading • **WILL NOT** ○		

- Subtasks
  - Warning message
  - Highlight insufficient mark

<table>
<tr><td rowspan="2">Story/Task name</td><td rowspan="2">Teacher Grade Viewing</td><td>ID</td><td>5</td></tr>
<tr><td>Size</td><td></td></tr>
<tr><td rowspan="3">Description of Customer benefits</td><td colspan="3">As a: Teacher</td></tr>
<tr><td colspan="3">I want: an overview of the grades I give to students</td></tr>
<tr><td colspan="3">So that: I can easily edit them if needed</td></tr>
<tr><td>Acceptance Criteria</td><td colspan="3">• MUST
○ Have a list of their classes with their students
○ Be able to view the grades of students in general
• SHOULD
○
• COULD
○
• WILL NOT
○</td></tr>
</table>

Story/Task name	Class Setup	ID	6
		Size	
Description of Customer benefits	As a: **Teacher**		
	I want: to be able to configure the class		
	So that: I can add teachers and students into it		
Acceptance Criteria	• **MUST** ○ Have a list of students at the school the teacher can add in ○ Add in teachers to the class ○ Create subjects for the class ○ Assign teachers to the subjects • **SHOULD** ○ Be able to add multiple students at the same time • **COULD** ○ Notify the student that they've been added • **WILL NOT** ○		

- Subtask
  - Create class
  - By creating class become the main teacher (regent)
  - Add other teachers after creating subject and assign them

Story/Task name	Account Login	ID	7
		Size	
Description of Customer benefits	As a: **User**		
	I want: to be able to login into the website		
	So that: I can access my necessary utilities		
Acceptance Criteria	• **MUST** ○ The users need to be stored to be accessed ○ Password must be securely stored and hashed • **SHOULD** ○ Show an error on failed login • **COULD** ○ • **WILL NOT** ○		

Subtasks:

- Create a login screen
- Link with database to login
- Create check with encryption
- Error handling

Story/Task name	Notification System	ID	8
		Size	
Description of Customer benefits	As a: **Student/Parent**		
	I want: to be notified on the website if change happened		
	So that: I can be up to date		

Acceptance Criteria	• **MUST**
	  ○ Check for updates on grades
	  ○ Have notification button on website
	• **SHOULD**
	  ○ Number of notifications
	• **COULD**
	  ○ Have separation of test creation or graded
	  ○ Indication of subject
	• **WILL NOT**
	  ○ Have a pop-up notification		

Subtasks:

- Create a notification box on websites
- Checks if teachers had done any modifications on last login
- Functionality of clearing notifcations

<table>
<tr><td rowspan="2">Story/Task name</td><td rowspan="2">Account creation</td><td>ID</td><td>9</td></tr>
<tr><td>Size</td><td></td></tr>
<tr><td rowspan="3">Description of Customer benefits</td><td colspan="3">As a: User</td></tr>
<tr><td colspan="3">I want: to create an account for the website</td></tr>
<tr><td colspan="3">So that: I can access the website</td></tr>
<tr><td>Acceptance Criteria</td><td colspan="3">• MUST
○ Account registration button
○ Account gets stored in the database
○ Unique username verification
○ Option for creating Student or Teacher account
• SHOULD
○ Password confirmation
○ Ask for a display name
○ Error if username already exists
• COULD
○
• WILL NOT
○ E-Mail link</td></tr>
</table>

Subtasks:

- Design login & registration screen
- Code the hashing of the password
- Create link with database to store login information

Story/Task name	Average grades	ID	10
		Size	
Description of Customer benefits	As a: **Student**		
	I want: to see a total average of my grades		
	So that: I know how I'm doing in my year		
Acceptance Criteria	• **MUST** ○ Calculate the average of the grades using the subjects coefficient ○ Display the average of the se-/trimester • **SHOULD** ○ Display the coefficients of the subjects for students ○ Display the averages accordingly • **COULD** ○ Display the average needed to compensate • **WILL NOT**		

Subtasks

- Allow teachers to add a coefficient for the subjects
- Display the average score on the page of the year

<table>
<tr><td rowspan="2">Story/Task name</td><td rowspan="2">Website Design</td><td>ID</td><td>11</td></tr>
<tr><td>Size</td><td></td></tr>
<tr><td rowspan="3">Description of Customer benefits</td><td colspan="3">As a: User</td></tr>
<tr><td colspan="3">I want: to be able to view the webpage</td></tr>
<tr><td colspan="3">So that: I can navigate through and access what I need</td></tr>
<tr><td>Acceptance Criteria</td><td colspan="3">• MUST
○ Have a clear design
○ Be navigable through buttons indicating what they do
• SHOULD
○ Have things stick out that are important
• COULD
○ Use a good colour scheme
○ Use design principles (Whitespace, ...)
• WILL NOT
○ Have accessibility options</td></tr>
</table>

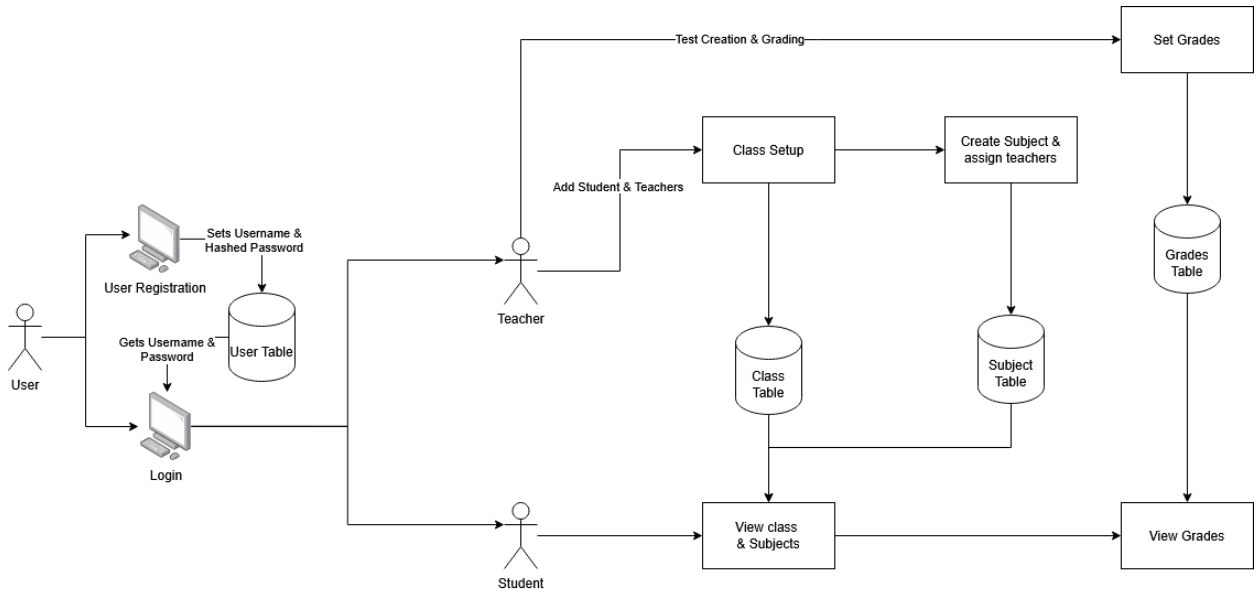
Subtasks:

- Create a design
- Implementation of web pages

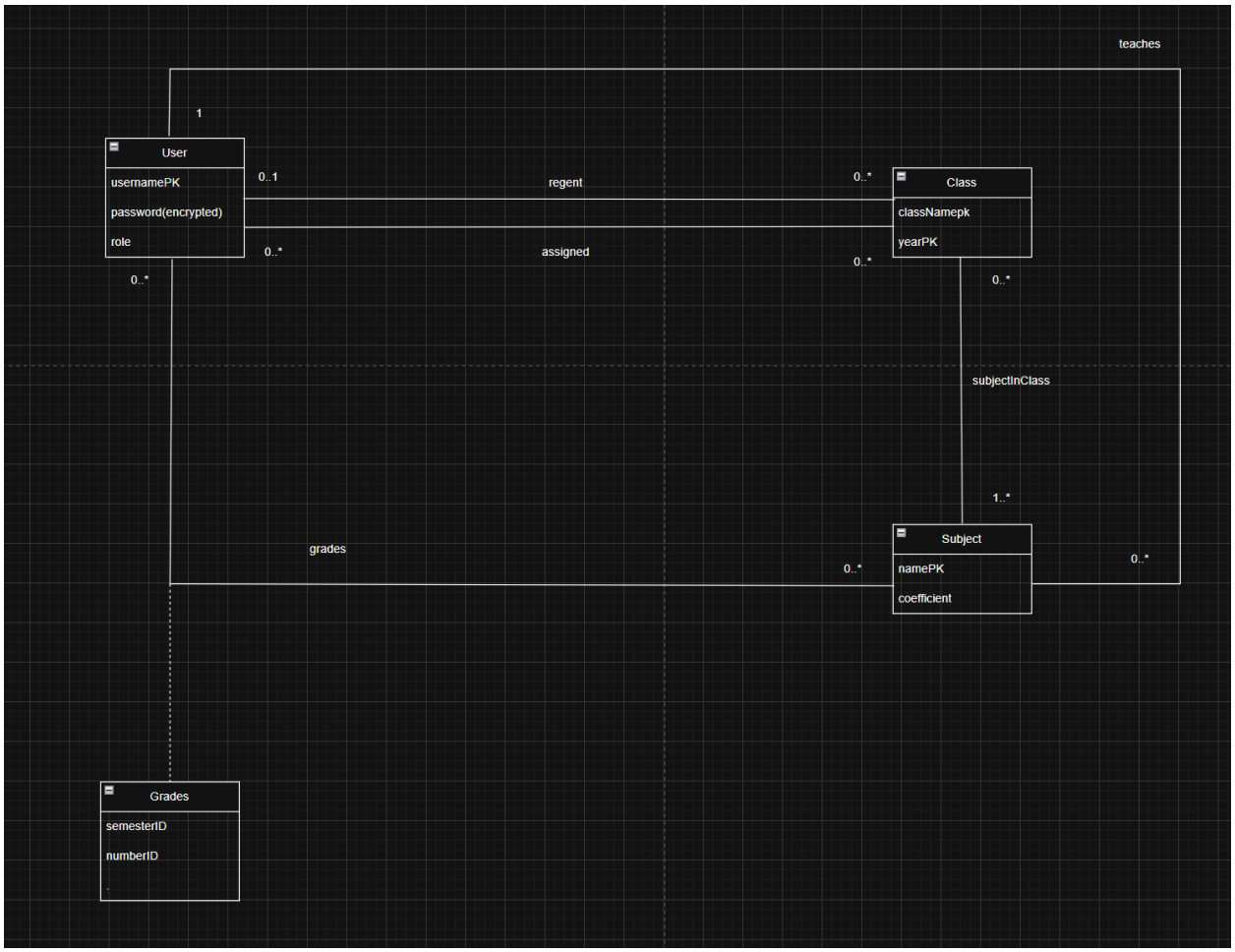
Story/Task name	Grade History	ID	12
		Size	
Description of Customer benefits	As a: **Student**		
	I want: to have a look at my old grades		
	So that: I can see my progress over the years		

Acceptance Criteria	• **MUST**
	  ○ Drop down menu for the classes
	• **SHOULD**
	  ○ Average **grade**
	• **COULD**
	  ○ Graph for the grade evolution
	• **WILL NOT**		

System Architecture Sketch



ER Diagram


teaches
1
User
usernamePK
password(encrypted)
role
0..1
regent
0..*
Class
classNamepk
yearPK
0..*
assigned
0..*
0..*
0..*
subjectInClass
1..*
Subject
namePK
coefficient
grades
0..*
0..*
Grades
semesterID
numberID